Re-analysis Summary Report for Peer-evaluation

Paper ID: **Anderson_AmEcoJourn_2011_bLe8**

Re-analyst's ID: **muyv3**

Your task

As a peer-evaluator, for this paper you are asked to judge whether the analyst's applied analytical choices for Task 1 and Task 2 are acceptable. By acceptable, we mean that the analysis pipeline is within the variations that could be considered appropriate by the scientific community in addressing the given analytical task. In addition, for Task 1, we ask you to judge whether the reported conclusion adequately follows from the results of the analysis; whether the co-analyst's self-categorization of the result is adequate. Although, you are not required, it would be appreciated, if you could execute the accompanying analysis code to check any mismatch between the results of the analysis and the reported results.

For both tasks, you should provide your evaluation via this survey: https://forms.gle/EAREaC6JjN2PWFfR6

The re-analyst's task

The re-analyst was instructed to assess the following claim of the original authors:
… [the study] finds substantially higher income for low-caste households residing in villages dominated by BACs [compared to villages dominated by upper caste]. (p. 240.)

The corresponding article

Here you can find the original article: https://osf.io/hcxdq/files/osfstorage/5e56aac535d22d02f75f482c?view_only=32ccf80d2c684ea9ac10ebbc4ec34df6
and the original data: https://osf.io/k7xzy/?view_only=c1f2b9aaf05d486c942783efd8ec57ec

They are provided only as context of the re-analysis.

In Task 1, the analyst was asked to conduct the analysis without any restrictions. Here, the analyst reported the following steps and results of the analysis:
I am giving a shorter summary of the analysis procedure here, please refer to the Analysis_MUYV3.html file in the Outputs folder for a full description of the methods and results, which includes figures, code, and a variable code book. In order to replicate the analyses, simply open the RProject file (Task1_MUYV3.RProj), then open the Markdown file in the Script folder (Analysis_MUYV3.Rmd). After installing the necessary packages, click knit to reproduce the analyses.
In order to test the claim, several key variables need to be operationalized first: lower caste households will be those from BAC, OBC, and SC castes, and all analyses will be restricted to those households. Village type will be operationalized through economic power, if the majority of the land is owned by high castes, the village is "high caste dominated". If the majority of the land is owned by low castes, the village is "low caste dominated". Income will be operationalized as the total household income, since the claim does not specify the source or type of income to be tested. Preprocessing: First, I removed all households with missing data. Second, dummy coded variables in the data set were combined into a single variable. Third, variables measuring "crop control" were summarized using their mean score since they measured the same construct

(Cronbach's alpha = 0.73). This reduced the number of possible predictors and simplified statistical models. Analysis 1: The analysis was performed in two parts. First, I tested whether high and low caste dominated villages differed only in caste dominance or whether they also differed in other respects (e.g. total area, type of crop grown, distance to infrastructure), which might confound village type comparisons. Using Mann Whitney U tests I found that high and low caste dominated villages only differed in caste dominance and not in any other control variable tested here. Analysis 2: I then assessed whether village level caste dominance affected household incomes of low caste households. For this purpose, I first removed outliers from the income data, and found that income was not normally distributed and was better approximated by a log-normal distribution. In order to test the effect of caste dominance on household income, I therefore estimated a multilevel generalized linear mixed model using a log-normal distribution, and modelling village, district, and state as nested random effects. This model was fitted using penalized quasi-likelihood estimation. Village caste dominance as well as a number of control variables (e.g. groundwater access, literacy) were added as fixed effects. The analysis showed that village caste dominance had no effect on household incomes $t(26) = 0.40$, $p = 0.69$.

The analyst's conclusion: The claim is not supported by the data.

The analyst's categorization of the result:
The results do not show evidence for or against the relationship/effect as described in the claim provided in your task

In Task 2, the analysts received the following instruction to conduct the analysis:
You should not use District controls, Crop controls, Distance controls, Groundwater controls, Public goods controls Do not include Irrigation, credit, and tenancy variables, and their interaction term versions Do not use instruments for water buyer status.

Here, the analyst reported the following steps and results of the analysis:
I am giving a shorter summary of the analysis procedure here, please refer to the Analysis_MUYV3.html file in the Outputs folder for a full description of the methods and results, which includes figures, code, and a variable code book. In order to replicate the analyses, simply open the RProject file (Task2_MUYV3.RProj), then open the Markdown file in the Script folder (Analysis_MUYV3.Rmd). After installing the necessary packages, click knit to reproduce the analyses. The analysis is highly similar to the one reported for Task 1, with the main difference being in the multilevel model estimation in Analysis 2. In order to test the claim, several key variables need to be operationalized first: lower caste households will be those from BAC, OBC, and SC castes, and all analyses will be restricted to those households. Village type will be operationalized through economic power, if the majority of the land is owned by high castes, the village is "high caste dominated". If the majority of the land is owned by low castes, the village is "low caste dominated". Income will be operationalized as the total household income, since the claim does not specify the source or type of income to be tested. Analysis 1: The analysis was performed in two parts. First, I tested whether high and low caste dominated villages differed only in caste dominance or whether they also differed in other respects (e.g. total area, type of crop grown, distance to infrastructure), which might confound village type comparisons. Using Mann Whitney U tests I found that high and low caste dominated villages only differed in caste dominance and not in any other control variable tested here. Analysis 2: I then assessed whether village level caste dominance affected household incomes of low caste households. For this purpose, I first removed households with missing data, and then removed outliers from the income data, and found that income was not normally distributed and was better approximated by a log-normal distribution. In order to test the effect of caste dominance on household income, I therefore estimated a multilevel generalized linear mixed model using a log-normal distribution, modelling village and state as nested random effects. This model was fitted using penalized quasi-likelihood estimation. Village caste dominance, three control variables (literacy, caste type, total village land), as well as their interactions were added as fixed effects. The analysis showed that village caste dominance had no effect on household incomes $t(66) = -0.872$, $p = 0.386$.

Analysis language/software/code (optional to check):

Task 1: R
Task 2: R
Analysis codes: https://osf.io/d4yk5/